

Claudin 1, 4, 6 and 18 isoform 2 as targets for the treatment of cancer (Review).

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The 24 claudin (CLDN) genes in the human genome encode 26 representative CLDN family proteins. CLDNs are tetraspan-transmembrane proteins at tight junctions. Because several CLDN isoforms, such as CLDN6 and CLDN18.2, are specifically upregulated in human cancer, CLDN-targeting monoclonal antibodies (mAbs), antibody-drug conjugates (ADCs), bispecific antibodies (bsAbs) and chimeric antigen receptor (CAR) T cells have been developed. In the present review, CLDN1-, 4-, 6- and 18.2-targeting investigational drugs in clinical trials are discussed. CLDN18.2-directed therapy for patients with gastric and other types of cancer is the most advanced area in this field. The mouse/human chimeric anti-CLDN18.2 mAb zolbetuximab has a single-agent objective response rate (ORR) of 9%, and increases progression-free and overall survival in combination with chemotherapy. The human/humanized anti-CLDN18.2 mAb osemitamab, and ADCs AZD0901, IBI343 and LM-302, with single-agent ORRs of 28-60%, have been tested in phase III clinical trials. In addition, bsAbs, CAR T cells and their derivatives targeting CLDN4, 6 or 18.2 are in phase I and/or II clinical trials. AZD0901, IBI343, zolbetuximab and the anti-CLDN1 mAb ALE.C04 have been granted fast track designation or priority review designation by the US Food and Drug Administration.

Targeting Claudin-18.2 for cancer therapy: updates from 2024 ASCO annual meeting.

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Multiple classes of therapies targeting claudin-18 isoform 2 (CLDN18.2) are under development for the treatment of advanced gastroesophageal adenocarcinoma and other solid tumors. At the 2024 American Society of Clinical Oncology (ASCO) Annual Meeting, the final results of the phase 3 SPOTLIGHT trial were presented, demonstrating a significant survival benefit from the addition of the CLDN18.2-specific antibody zolbetuximab to chemotherapy in the first-line treatment of advanced gastroesophageal adenocarcinomas with $\geq 75\%$ CLDN18.2 expression. Early-phase trial results presented at ASCO 2024 showed promising efficacy and safety of the afucosylated CLDN18.2-specific antibody FG-M108 in combination with chemotherapy in the first-line treatment of CLDN18.2-positive advanced gastroesophageal and pancreatic cancers. In addition, several early-phase trials presented at ASCO 2024 investigate other CLDN18.2-targeting approaches in CLDN18.2-positive refractory advanced solid tumors, including the CLDN18.2-targeting antibody-drug conjugates LM-302 and IBI343, the bispecific anti-CLDN18.2/CD3 antibody IBI38, and the chimeric antigen receptor T cell therapy satricabtagene autoleucel. These novel approaches could potentially expand the benefit of CLDN18.2-targeting therapies to a broader range of tumor types and to tumors expressing lower levels of CLDN18.2.

Zolbetuximab plus mFOLFOX6 in patients with CLDN18.2-positive, HER2-negative, untreated, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma (SPOTLIGHT): a multicentre, randomised, double-blind, phase 3 trial.

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Zolbetuximab, a monoclonal antibody targeting claudin-18 isoform 2 (CLDN18.2), has shown efficacy in

patients with CLDN18.2-positive, human epidermal growth factor receptor 2 (HER2)-negative, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma. We report the results of the SPOTLIGHT trial, which investigated the efficacy and safety of first-line zolbetuximab plus mFOLFOX6 (modified folinic acid [or levofolinate], fluorouracil, and oxaliplatin regimen) versus placebo plus mFOLFOX6 in patients with CLDN18.2-positive, HER2-negative, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma. SPOTLIGHT is a global, randomised, placebo-controlled, double-blind, phase 3 trial that enrolled patients from 215 centres in 20 countries. Eligible patients were aged 18 years or older with CLDN18.2-positive (defined as $\geq 75\%$ of tumour cells showing moderate-to-strong membranous CLDN18 staining), HER2-negative (based on local or central evaluation), previously untreated, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma, with radiologically evaluable disease (measurable or non-measurable) according to Response Evaluation Criteria in Solid Tumors version 1.1; an Eastern Cooperative Oncology Group performance status score of 0 or 1; and adequate organ function. Patients were randomly assigned (1:1) via interactive response technology and stratified according to region, number of organs with metastases, and previous gastrectomy. Patients received zolbetuximab (800 mg/m² loading dose followed by 600 mg/m² every 3 weeks) plus mFOLFOX6 (every 2 weeks) or placebo plus mFOLFOX6. The primary endpoint was progression-free survival assessed by independent review committee in all randomly assigned patients. Safety was assessed in all treated patients. The study is registered with ClinicalTrials.gov, NCT03504397, and is closed to new participants. Between June 21, 2018, and April 1, 2022, 565 patients were randomly assigned to receive either zolbetuximab plus mFOLFOX6 (283 patients; the zolbetuximab group) or placebo plus mFOLFOX6 (282 patients; the placebo group). At least one dose of treatment was administered to 279 (99%) of 283 patients in the zolbetuximab group and 278 (99%) of 282 patients in the placebo group. In the zolbetuximab group, 176 (62%) patients were male and 107 (38%) were female. In the placebo group, 175 (62%) patients were male and 107 (38%) were female. The median follow-up duration for progression-free survival was 12.94 months in the zolbetuximab group versus 12.65 months in the placebo group. Zolbetuximab treatment showed a significant reduction in the risk of disease progression or death compared with placebo (hazard ratio [HR] 0.75, 95% CI 0.60-0.94; $p=0.0066$). The median progression-free survival was 10.61 months (95% CI 8.90-12.48) in the zolbetuximab group versus 8.67 months (8.21-10.28) in the placebo group. Zolbetuximab treatment also showed a significant reduction in the risk of death versus placebo (HR 0.75, 95% CI 0.60-0.94; $p=0.0053$). Treatment-emergent grade 3 or worse adverse events occurred in 242 (87%) of 279 patients in the zolbetuximab group versus 216 (78%) of 278 patients in the placebo group. The most common grade 3 or worse adverse events were nausea, vomiting, and decreased appetite. Treatment-related deaths occurred in five (2%) patients in the zolbetuximab group versus four (1%) patients in the placebo group. No new safety signals were identified. Targeting CLDN18.2 with zolbetuximab significantly prolonged progression-free survival and overall survival when combined with mFOLFOX6 versus placebo plus mFOLFOX6 in patients with CLDN18.2-positive, HER2-negative, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma. Zolbetuximab plus mFOLFOX6 might represent a new first-line treatment in these patients. Astellas Pharma, Inc.

Claudin 18.2 expression in various tumor types and its role as a potential target in advanced gastric cancer.

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Alterations in claudin expression can impair tight junction function, influence signaling pathways, and act

as a tumor-promoting event in some epithelial cancers. Recently, zolbetuximab, a highly potent and tumor cell-selective therapeutic antibody against claudin 18.2, has been developed and investigated in clinical trials. We conducted a prospective study using claudin 18.2 immunohistochemistry in 430 consecutive patients with advanced gastrointestinal, genitourinary, or rare cancers between June 2012 and March 2016. Claudin 18.2 expression was evaluated in 96.3% of the patients (414/430) using immunohistochemistry. In total, 4.1% (17/414) of the patients were claudin 18.2-positive, including patients with pancreatic (16.7%, 1/6), gastric (14.1%, 12/85), biliary tract (6.3%, 1/16), genitourinary/miscellaneous (2.2%, 1/46), and colorectal (0.9%, 2/203) cancers. Twelve of 17 patients positive for claudin 18.2 had gastric cancers (GCs); this subgroup showed no statistical differences by gender, age, disease extent, primary tumor site, pathologic differentiation, human epidermal growth factor receptor 2, or Epstein-Barr virus status with or without claudin 18.2 expression. However, claudin 18.2 was more frequently positive in intestinal-type compared with diffuse-type as assessed by Lauren classification ($P=0.026$). There was no significant difference in overall survival (OS) between patients with and without claudin 18.2 expression ($P=0.101$). Our results add to the emerging literature about claudin 18.2 expression in various cancer types and support the need for extended clinical exploration of zolbetuximab.

[Claudine 18.2: A new therapeutic target in digestive cancers].

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Therapies targeting HER2 and immune checkpoint inhibitors have improved survival in patients with metastatic gastric or gastro-oesophageal junction adenocarcinoma, but the prognosis associated with these cancers remains poor. Claudin 18.2 is a tight junction protein expressed in the oeso-gastric mucosa. Some gastric and gastro-oesophageal junction adenocarcinoma overexpress this protein, as well as some pancreatic, ovarian and pulmonary cancers. In pathological context, its epitope may be exposed at the surface of cells and therefore makes it an interesting therapeutic target. Zolbetuximab, a monoclonal antibody targeting claudin 18.2, showed a survival benefit in first line metastatic treatment in patients with claudin 18.2 positive gastric and gastro-oesophageal junction adenocarcinoma, in two phase III studies. CAR T-cells specifically targeting this protein have also shown promising efficacy from the second line of treatment. Considering the probable impact of the expression status of claudin 18.2 in future treatment algorithms, this review aims to present the pathophysiology underlying the targeting of claudin 18.2, summarize state of the art results of anti-claudin 18.2 therapies and discuss future challenges for the management of patients with claudin 18.2 positive gastric and gastro-oesophageal junction adenocarcinoma.

A multicentre, phase IIa study of zolbetuximab as a single agent in patients with recurrent or refractory advanced adenocarcinoma of the stomach or lower oesophagus: the MONO study.

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Claudin 18.2 (CLDN18.2) is physiologically confined to gastric mucosa tight junctions; however, upon malignant transformation, perturbations in cell polarity lead to CLDN18.2 epitopes being exposed on the cancer cell surface. The first-in-class monoclonal antibody, zolbetuximab (formerly known as IMAB362),

binds to CLDN18.2 and can induce immune-mediated lysis of CLDN18.2-positive cells. Patients with advanced gastric, gastro-oesophageal junction (GEJ) or oesophageal adenocarcinomas with moderate-to-strong CLDN18.2 expression in $\geq 50\%$ of tumour cells received zolbetuximab intravenously every 2 weeks for five planned infusions. At least three patients were enrolled in two sequential cohorts (cohort 1300 mg/m²; cohort 2600 mg/m²); additional patients were enrolled into a dose-expansion cohort (cohort 3600 mg/m²). The primary end point was the objective response rate [ORR: complete and partial response (PR)]; secondary end points included clinical benefit [ORR+stable disease (SD)], progression-free survival, safety/tolerability, and zolbetuximab pharmacokinetic profile. From September 2010 to September 2012, 54 patients were enrolled (cohort 1, n = 4; cohort 2, n = 6; cohort 3, n = 44). Three patients in cohort 1 and 25 patients in cohorts 2/3 received at least 5 infusions. Antitumour activity data were available for 43 patients, of whom 4 achieved PR (ORR 9%) and 6 (14%) had SD for a clinical benefit rate of 23%. In a subgroup of patients with moderate-to-high CLDN18.2 expression in $\geq 70\%$ of tumour cells, ORR was 14% (n = 4/29). Treatment-related adverse events occurred in 81.5% (n = 44/54) patients; nausea (61%), vomiting (50%), and fatigue (22%) were the most frequent. Zolbetuximab monotherapy was well tolerated and exhibited antitumour activity in patients with CLDN18.2-positive advanced gastric or GEJ adenocarcinomas, with response rates similar to those reported for single-agent targeted agents in gastric/GEJ cancer trials. NCT01197885.

Claudin18.2 in Advanced Gastric Cancer.

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Globally, the fifth most common cancer and the fourth leading cause of cancer-related mortality is gastric cancer (GC). Recent clinical trials on solid tumors enrolled patients who possess druggable genetic alterations, protein expression, and immune characteristics. In gastric or gastroesophageal junction (GEJ) cancers, trastuzumab combined with first-line chemotherapy in human epidermal growth factor receptor 2 (HER2)-positive patients and ramucirumab combined with second-line paclitaxel remarkably prolonged overall survival (OS) compared with chemotherapy alone, according to phase 3 trial results. Recently, immune checkpoint inhibitor (ICI) monotherapy was approved as third- or later-line treatment. Chemotherapy plus ICIs as first-line treatment exhibited improved survival compared with chemotherapy alone in HER2-negative patients according to Checkmate 649 trial results. Conversely, systemic chemotherapy prognosis remains poor, although some patients may achieve durable response to treatment and prolonged survival in advanced GC. Recently, a first-in-class, chimeric immunoglobulin G1 monoclonal antibody (zolbetuximab) that targets and binds to claudin 18 isoform 2 (CLDN18.2) has emerged as a new target therapy in GC treatment. Global phase III trials revealed that the addition of zolbetuximab to first-line chemotherapy prolonged OS in CLDN18.2-positive and HER2-negative GC patients. This review summarizes recent clinical trials of CLDN18.2-targeted therapy.

Premenstrual syndrome, a common but underrated entity: review of the clinical literature.

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Premenstrual syndrome (PMS) and premenstrual dysphoric disorder (PMDD) are characterized by somatic and psychologic symptoms that arise at the luteal phase of the menstrual cycle and subside with

menstruation. For definitive diagnosis prospectively self-reported symptoms should demonstrate a cyclic pattern and other psychological pathologies and thyroid dysfunction, that may present with similar symptoms, should be excluded. Both entities affect millions of women at reproductive age as the prevalence of PMS is given as 10-98% while PMDD affects 2-8%. Sex steroids and neurotransmitters have a central role in the etiology. The role of vitamins and minerals in the etiology and treatment of PMS and PMDD is open to discussion. Drugs that suppress ovarian sex steroid production, such as combined oral contraceptives or selective serotonin re-uptake inhibitors enhancing central serotonin delivery are used for treatment. Life-style changes and regular exercise also have a positive effect in milder cases. Tricyclic antidepressants and gonadotropin-releasing hormone analogues can be used in selected cases.

Prevalence and Correlations of Premenstrual Dysphoric Disorder: A Sample of Adult Working Women.

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Premenstrual dysphoric disorder (PMDD) represents a significant psychiatric condition affecting women of reproductive age. It is characterized by cyclical emotional and physical symptoms that considerably impair psychosocial functioning. This study aims to investigate the prevalence of PMDD among adult working women and identify associated etiological factors. From January 2022 to March 2022, a cross-sectional study utilizing cluster sampling methodology was conducted to investigate the prevalence of PMDD among adult working women in six communities in Hangzhou (China). A total of 1710 participants were selected for research through a questionnaire-based survey. To elucidate the factors associated with PMDD, logistic regression analysis was employed. Among 1710 working women, 52 had PMDD, representing an incidence of 3.04%. Age (odds ratio (OR) = 3.196, $p = 0.002$), age at menarche (OR = 0.362, $p = 0.011$), income (OR = 0.449, $p = 0.043$), menstrual cycle (OR = 2.352, $p = 0.022$), menstrual symptoms (OR = 0.317, $p = 0.003$), physical activity (OR = 2.234, $p = 0.028$), and caffeine intake (OR = 2.318, $p = 0.022$) were independent influencing factors of PMDD in adult working women. The study highlights that PMDD, which has a prevalence of 3.04% in the surveyed population, is a notable concern among adult working women. Key factors, such as age, age at menarche, income level, menstrual cycle characteristics, menstrual symptom presence, physical activity, and caffeine intake, significantly contribute to the likelihood of developing PMDD. These findings underscore the importance of recognizing and monitoring the above risk factors by clinicians to identify and manage PMDD in working women.

The prevalence, impairment, impact, and burden of premenstrual dysphoric disorder (PMS/PMDD).

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Currently it is estimated that 3-8% of women of reproductive age meet strict criteria for premenstrual dysphoric disorder (PMDD). Assessment of published reports demonstrate that the prevalence of clinically relevant dysphoric premenstrual disorder is probably higher. 13-18% of women of reproductive age may have premenstrual dysphoric symptoms severe enough to induce impairment and distress, though the number of symptoms may not meet the arbitrary count of 5 symptoms on the PMDD list. The impairment and lowered quality of life for PMDD is similar to that of dysthymic disorder and is not much lower than

major depressive disorder. Nevertheless, PMS/PMDD is still under-recognized in large published epidemiological studies, as well as assessments of burden of disease. It is demonstrated here that the burden of PMS/PMDD as well as the disability adjusted life years (DALY) lost due to this repeated-cyclic disorder is in the same magnitude as major recognized disorders. Appropriate recognition of the disorder and its impact should lead to treatment of more women with PMS/PMDD. Efficacious treatments are available. They should reduce individual suffering and impact on family, society, and economy.
